

C Programming

Trainer :Rajiv K

Email :rajiv.kamune@sunbeaminfo.com



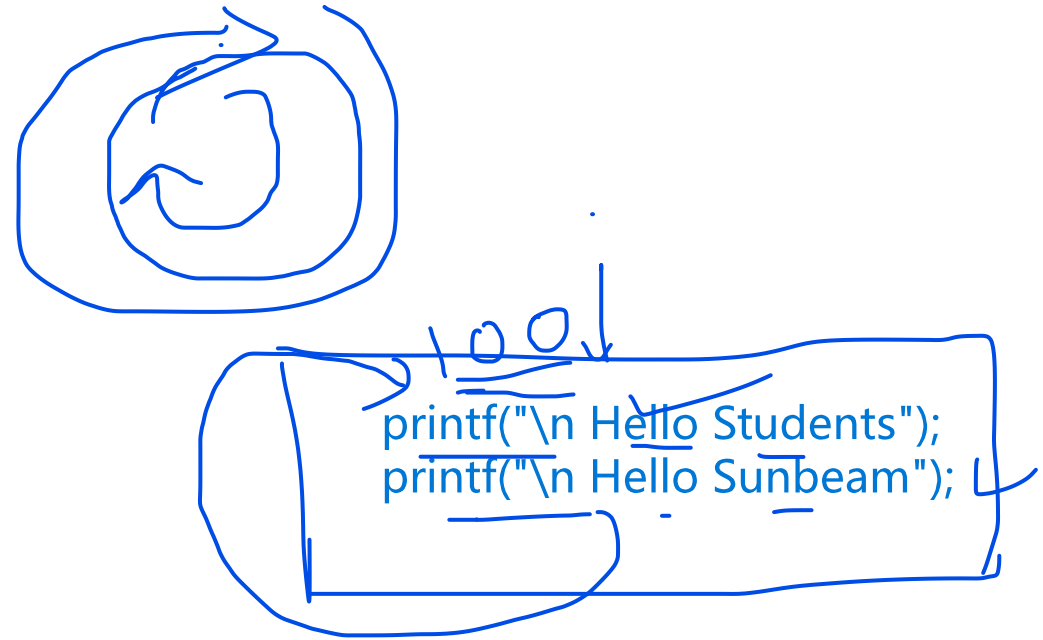
Agenda

- Loops
- For loop
- While loop
- Do... while loop
- Infinite loop
- Jump Statements
- break, continue
- return, goto



Loops - Introduction :

- Control statements used for repeating a set of instructions number of times is called as "LOOP".
- Loops help us to execute a statement /block of statements a certain number of times.
- Loops can be categorized as :
 - 1) Entry Controlled Loops
 - 1) While Loop
 - 2) For Loop
 - 2) Exit Controlled Loops
 - 1) Do - While Loop
- Every loop has
 1. Initialization statement
 2. Terminating condition
 3. Modification statement(Increment/Decrement)
 4. Body of loop
- The variable that is used for terminating condition is referred as "loop/control variable".



Entry Controlled Loops :

- Checks the condition at an entry level(at the beginning).
- Execution Flow :
 - 1) The control Variable is **initialized** first.
 - 2) The variable is evaluated with the **condition**.
 - 3) If the condition is **false**, the loop is **terminated**, else
 - 4) If **true**, the loop **body is executed** and the variable is **updated**(incremented/decremented).
 - 5) Steps 2 and 4 are **repeated until condition is false**.
- Types of Entry controlled Loops:
 - 1) **While Loop**
 - 2) **For Loop**



While Loop :

Syntax :

- Control variable initialization
- While (condition) // (control statement)
{
 // Loop Body
 control variable increment/decrement
}
- Execution control enters the block ONLY if CONDITION IS TRUE.



For Loop :

Syntax :

• **For**(variable initialization; **condition**; modification)

{
loop Body;

}

• Execution control enters the block ONLY if CONDITION IS TRUE.

1 false
int i=1; true

*i < 5 ; 1 < 5 => 1
2 < 5 => 1
3 < 5 => 1*

i++ 1+1=>2+1=>3 5

5 < 5 => 0

*printf("\n %d", 2*i); 2*1=> 2
2*2=>4*

loop terminated



Exit Controlled Loop :

- Checks the condition at an exit level(at the end).
- Loop body executes at least once irrespective of the condition being true or false.
- Execution Flow :
 - 1) The control variable is initialized first.
 - 2) The loop body is executed. The variable gets updated(incremented/decremented).
 - 3) The variable is evaluated with the condition.
 - 4) If condition is false loop gets terminated else
 - 5) Steps 2 and 3 are repeated till the condition is false.
- Type of Exit Controlled Loop
 - 1) Do - while Loop



Do - While Loop

Syntax :

- Variable initialization

Do

```
{  
    loop Body;  
    variable modification  
} while (condition);
```

- Loop executes at least once irrespective of the condition.



Example :

while

```
int num = 5;  
while (num <= 3)  
{  
    printf("Inside Loop ");  
}
```

In this case, the control DOES NOT enter the loop as the condition is false at the entry level.

Do-while

```
int num = 5;  
do  
{  
    printf("inside Loop");  
}while(num<=3);
```

In this case, the control goes inside the loop at least once as the condition is checked at the exit.



Infinite loop :

- If loop condition is always true, program never terminates.
- `while(1) {`
 ...
 }
- `for(;;) {`
 ...
 }
- `do {`
 ...
} `while(1);`



Jump statements : break , continue :

- Break :

- Used to early exit from loop, or to exit an infinite loop
- Takes control out of current loop and continues execution of statements after the loop.
- Statements after break are skipped.
- break is used with loop/switch case.
- In case of nested loops, break affects current loop only (not outer).

- Continue :

- Used to continue next iteration of the loop.
- Statements after continue are skipped (for current iteration).
- Continue is used only with loops.
- In case of nested loops, continue affects current loop only (not outer).



Jump statements : return , goto :

- Return :
 - Can be used inside function.
 - Helps to move execution control forcefully back to calling function.
- goto :
 - Jumps to statement label, must be within same function as the goto.
 - Statement label is an identifier followed by a colon (:)
 - Unstructured control statement
 - Used rarely (less readable)
 - Advised to use only for forward jump
 - Best use is to exit from deeply nested loops.
 - Syntax:
 - goto label name;
 -
 -
 -
 - label_name: C-statements



Thank You

